Derek Liu

dxl540.github.io
derekl@mit.edu | +1 (858) 733-1018

EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Double Major in Mathematics (Course 18) and Computer Science (Course 6-3)
2023-2027
GPA: 5.0/5.0

HONORS AND AWARDS

2023-2025	Putnam Top 16 Rank (N1)
2023	64th International Math Olympiad (IMO) Gold Medalist (Perfect Score)
2023	USA Math Olympiad (USAMO) Gold Award (Perfect Score)
2023	Regeneron Science Talent Search (STS) Scholar
2022	63rd International Math Olympiad (IMO) Gold Medalist
2022	USA Math Olympiad (USAMO) Gold Award

RESEARCH AND WORK EXPERIENCE

SUMMER PROGRAM IN UNDERGRADUATE RESEARCH (SPUR)
Jun 2026 - Aug 2026 | Cambridge, MA
Summer research program in mathematics at MIT. My project was on interpolating Kac-Moody Lie algebras and their properties to non-integer parameters using Deligne categories. Partnered with Daniel Potievsky and mentored by Ivan Motorin.

FIVE RINGS | Quantitative Research Analyst Intern
January 2026 | New York, NY
4-week winter internship

MIT UROP | Paul E. Gray UROP Researcher
Aug 2025 - current | Cambridge, MA
Supervised by Professor Andrew Sutherland on solving index form equations in quintic number fields. Working on implementing an algorithm in Sage (GitHub), preparing to submit to LuCaNT 2027.

JANE STREET | Trading Intern
May 2025 - Aug 2025 | New York, NY
11-week summer internship

REU DULUTH Jun 2024 - Aug 2024 | Duluth, MN
Summer research program in combinatorics/number theory at UM Duluth

PRIMES USA Jan 2022 - Dec 2022
MIT-run research program for high school students, mentored by Yuan Yao

SERVICE

OMEGA | Creative Team Member
Jan 2026 - current
OMEGA is an organization running olympiad-style math contests across the nation. Members of the Creative Team are responsible for all academic matters, including creating, selecting, and proofreading problems for tests.

NEXT SING | Show Co-Coordinator, Co-President
Feb 2025 - current
Next Sing is an all-levels a cappella group for MIT students. As co-president in 2025, I ran events and executive meetings. As show co-coordinator in 2026, I oversaw show logistics such as room bookings and concert brochures.

HMMT | Historian, Problems Advisor, Problem Czar
April 2024 - current
HMMT is a team math competition run by Harvard and MIT students. As Historian from 2026-2027, I help propose projects, oversee logistics, and transition the organization between seasons. As a Problems Officer from 2024-2026, I was responsible for selecting staff and organizing problem-writing meetings to assemble the tests.

18.A34 (PUTNAM SEMINAR) | Guest Lecturer
Fall 2024, Fall 2025
Taught a lecture on linear algebra in combinatorics in both years, supervised by Professor Henry Cohn in 2025 and Professor Bjorn Poonen in 2024.

USA TST SERIES | Grader
Nov 2023 - Jan 2025
The USA Team Selection Test series is a series of tests used to select USA's IMO team. It includes the USA TST, RMM (Romanian Masters in Mathematics), APMO (Asian Pacific Math Olympiad), and USAMO. I was a grader for both the 2023-24 and 2024-25 cycles.

18.404 (THEORY OF COMPUTATION) | Grader
Fall 2024, Fall 2026

PUBLICATIONS

Colin Defant and Derek Liu. Homology in combinatorial refractive billiards. *Res Math Sci*, 12(2):36, 2025. doi:10.1007/s40687-025-00524-8.

Derek Liu and Yuan Yao. Arrangements of simplices in fine mixed subdivisions. 2022. https://math.mit.edu/research/highschool/primes/materials/2022/Liu-Yao.pdf.